



Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression

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ARTICLE INFO

Keywords:

Vortex-induced vibration
Long-span bridge
Field measurement
Data-driven approach
Machine learning

ABSTRACT

In recent years, vortex-induced vibrations (VIVs) have been observed on a few long-span bridges. Although a number of studies of VIV on structures have been conducted, and semi-empirical models have been proposed based on wind tunnel experiments, it is very difficult to accurately simulate the VIVs of prototype full-scale structures because of the complicated associated wind conditions, the Reynolds number effect, three-dimensional (3D) aeroelastic effects, uncertainties in the dynamic structure model, etc. In this paper, a data-driven approach using a machine learning (ML) scheme is proposed to model VIV response based on a database of field-measured VIVs for a full-scale long-span suspension bridge over six recent years. First, a decision tree learning algorithm is adopted to train the VIV mode classification model by mapping the relationship between wind characteristics and VIV mode; this can be employed to automatically identify VIV modes. Next, a support vector regression (SVR) algorithm is employed to model the VIV response of the bridge deck in the time domain for separate VIV modes. In this ML scheme, the wind speeds and wind directions at three different locations along the bridge span are used as the exogenous inputs of the classification model and regression model, taking the nonuniformity of the wind field into account. For the regression model, the VIV response at a previous step is also involved as an additional input variable. The trained models are validated for a few measured VIV events, which are randomly selected from the field measurement database. Further, the parametric analysis of the VIV response is performed using the trained model. The results indicate that the classification model and regression model can, respectively, accurately identify and predict the VIV response for various modes of the investigated bridge. The wind speed, wind direction, and nonuniformity are the critical parameters for the overall process (developing stage, steady stage, and vanishing stage) and amplitude of the VIV response.

1. Introduction

As the span of modern bridges increases, the structures become more flexible and have slight damping. These long-span bridges are thus increasingly sensitive to crosswind loading caused by vortex shedding. Vortex-induced vibrations (VIVs) have been observed on a number of bridges, such as the Great Belt Bridge (Larsen et al., 2000; Frandsen, 2001), Tokyo Bay Bridge (Fujino and Yoshida, 2002), and Xihoumen Bridge (Li et al., 2011, 2014a). VIVs on long-span bridges commonly occur within a low wind-speed range because of the associated low structural frequencies. Accordingly, VIV occurs relatively frequently on long-span bridges, which may result in long-term fatigue damage, thus

shortening bridge life.

VIV is an aeroelastic phenomenon involving strong interactions between flow and structure. It occurs during periodic vortex shedding when the shedding frequency is identical to or approaches the natural structural frequency. Comprehensive investigations on the mechanisms of and modeling for VIV on bridges have been carried out. Nakamura and Mizota (1975) have measured the lifts and wakes of rectangular prisms with various aspect ratios oscillating transversely in a uniform flow, with the short sides normal to the flow direction in a wind tunnel. The “lock-in” phenomenon is observed for all models with different aspect ratios. It is found that the phase angles of the frequency response components of both the lift and near-wake velocity show abrupt changes

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<https://doi.org/10.1016/j.jweia.2017.10.022>

Received 21 February 2017; Received in revised form 23 October 2017; Accepted 23 October 2017

Available online 20 November 2017

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Nomenclature list

VIV	Vortex-induced vibration	N	Node number in decision tree
ML	Machine Learning	RMS	Root mean square
SVR	Support Vector Regression	A	RMS of vertical displacement
F	Aerodynamic force	$g(x)$	A function to be approximated
ρ	Air density	x_i	Input of a sample
U	Wind flow speed	y_i	Output (target value) of a sample
θ	Wind direction	M	Size of the training set
D	Across-flow dimension of the structure	Φ	Nonlinear mapping
Y_1, Y_2, Y_3	Aerodynamic derivatives	w	A vector of weights
ω	Vortex shedding frequency	b	Offset of the discriminating hyperplane
λ	A constant	C	Penalty parameter
min	Minute	ξ, ξ^*	Slack variables
y	Across-flow displacement of the structure	ϵ	A range parameter
$\tilde{C}_L$	Stochastic lift force coefficient	$\alpha, \alpha^*, \beta, \beta^*$	Lagrangian multipliers
VIVF	Vortex-induced vertical force	KKT	Karush-Kuhn-Tucker condition
NLSF	Nonlinear least square fitting	NMSE	Normalized mean square error
AI	Artificial intelligence	SCC	Squared correlation coefficient
ANN	Artificial neural network	γ	A parameter determining the radial range
CA	Cellular automata	$\bar{U}$	Average of mean wind speeds at 1/4, 1/2 and 3/4 center span
PSD	Power spectrum density	MinLeaf	Minimum number of samples in the leaf node
FEM	Finite element method	f_0	The function (model) reflecting the characteristics of the VIV dynamic system
f_s	Natural frequency of the structure	α_1, α_2	Fractions of training samples sent down to child nodes from the parent node
I	Gini impurity index		
P	Fraction of samples		
ω_i	i th VIV mode		

when the critical reduced wind velocity for vortex shedding is approached. This is suggested to be a key phenomenon involved in solving the problem of the vortex excitation of bluff structures. Komatsu and Kobayashi (1980) have conducted a series of experiments on various cross-sections (such as L-shaped, T-shaped, H-shaped, and rectangular cylinders) with various aspect ratios in a wind tunnel. Two types of VIVs are found. One is a forced small-amplitude vibration caused by Karman vortex-shedding in cylinders (T-cylinders) with a separation point at the trailing edge. The other is a self-excited vibration with a relatively large amplitude in cylinders (L-, H- and rectangular cylinders) with a separation point at the leading edge, which occurs independently of the Karman vortex. The generating mechanism in the latter case is described as a motion-induced vortex at the leading edge that synchronizes with the motion of the cylinder. The frequency of this type of vibration does not change within a certain range of wind velocities and coincides with the natural frequency of the cylinder, termed with “lock-in” phenomenon. Matsumoto et al. (1993) have investigated the hybrid characteristics of VIV caused by Karman vortex shedding and the motion-induced vortices for a rectangular cylinder with an aspect ratio of 4:1 and a hexagonal girder with/without handrails. The results indicate that VIVs for both models are essentially excited by motion-induced vortices, which are mitigated by the presence of Karman vortices. Laima et al. (2013) have employed a hybrid method combining experiments with numerical simulations to study the flow characteristics of a twin-box girder with a gap ratio of 1.7. The results indicate that strong vortices are generated in the gap when the response amplitude of VIV is large. Thus, to disturb the vortices for suppressing VIV, cross bars with different schemes and light flat plates are installed in the gap. A plate covering the gap is found to achieve the best suppression of VIV for this twin-box girder. Li et al. (2014b) have investigated the Reynolds number effects on the aerodynamic characteristics and VIV of a twin-box girder within a range of Reynolds number values ($5.85 \times 10^3 \sim 1.12 \times 10^5$). They find that the transition point of the separated shear layer moves upstream, and the bubble size gradually decreases with increasing Reynolds number values.

The VIV on a twin-box girder with a higher Reynolds number generally corresponds to a lower critical reduced wind velocity, larger vibration amplitude, and wider “lock-in” range.

VIV modeling is quite important in the design of structures (here, modeling refers to the model of aerodynamic forces acting on bridges). Rigorous mathematical-physical modeling of VIV requires simultaneously solving the Navier-Stokes (N-S) equation and motion of equation of the structure. However, because of the strong nonlinearity of the N-S equation, this has proved to be mathematically intractable (Basu and Vickery, 1983). As a less-than-ideal alternative, simplified semi-empirical models are proposed based on wind tunnel experiments. A wake oscillator system has been used to model VIV since the early work of Bishop and Hassan (1964). Specifically, VIV is modeled by a system of two coupled equations, where a second-order linear mechanical oscillator is used to model structural motion, and a second-order nonlinear wake oscillator is used to model vortex-induced exciting force (Hartlen and Currie, 1970). The parameters in the model are identified from experimental wind tunnel results. Simiu and Scanlan (1986) proposed a single ordinary differential equation to model VIV, and the parameter identification approach was then proposed by Ehsan and Scanlan (1990). In the Scanlan model, aerodynamic force is given by

$$F = \frac{1}{2} \rho U^2 (2D) \left[Y_1(K) \left(1 - \lambda \frac{y^2}{D^2} \right) \frac{\dot{y}}{U} + Y_2(K) \frac{y}{D} + \frac{1}{2} \tilde{C}_L(t) \right] \quad (1)$$

where D is the across-flow dimension of the body; ρ is the air density; $K = \omega D/U$ is the reduced frequency of vortex shedding, where ω is the vortex shedding frequency that satisfies the Strouhal relationship (outside “lock-in”), $\omega D/U = 2\pi S_t$ and S_t is the Strouhal number, which is related to the shape of the bluff body and the Reynolds number; λ is a constant denoting the nonlinear dependence of self-excited force on displacement amplitude; Y_1 and Y_2 are the aerodynamic derivatives; and $\tilde{C}_L$ is the stochastic lift force coefficient. The total force consists of two types of forces: one is induced by vortex shedding around the bluff body (the third term with $\tilde{C}_L$

in Eq. (1)) and the other is a motion-induced lift force (the first two terms in Eq. (1)), which includes an aerodynamic damping Y_1 and an aerodynamic stiffness Y_2 . The force induced by vortex shedding (the term with $\tilde{C}_L$) is very small compared to the motion-induced forces created by dramatic oscillations (Ehsan and Scanlan, 1990). In addition, the effects of the aerodynamic damping force and stiffness force on the aerodynamic force have also been studied. It is concluded that the aerodynamic damping is negative, and its absolute value decreases with the vibration amplitude of the structure in the wind tunnel experiment, implying an increase in total damping (Diana et al., 2006). From field measurements of a long-span suspension bridge, Li et al. (2014a) have found that total damping varies greatly with wind velocity, while total stiffness remains approximately constant, implying that the effect of aerodynamic stiffness force on VIV is very relatively weak compared to that of the aerodynamic damping force. Zhu et al. (2013) have investigated the nonlinear vortex-induced vertical force (VIVF) on a flat closed-box deck via wind tunnel tests of a large-scale (1:20) spring-suspended section model. They apply a nonlinear least square fitting (NLSF) method on the time histories of the VIVFs extracted from the measured dynamic force to identify the relevant parameters of the Scanlan empirical nonlinear model (1990), and they improve the Scanlan nonlinear model by adding a quadratic term $Y_3 \frac{\dot{y}}{UB}$ for the VIVFs. The results show that the improved nonlinear model is adequate, while the original Scanlan model may not be suitable for describing VIVF acting on a flat closed-box deck.

The above models are proposed based on wind tunnel experiments, and the parameters in those models are also derived from wind tunnel experiments. However, the discrepancy between the wind tunnel experiments and prototype bridge results in the inability of these models to accurately predict VIV responses for a prototype bridge: (1) the section model in the wind tunnel cannot reflect the three-dimensional aeroelastic characteristics for the VIV of the full-scale bridge; (2) the above models assume that wind is uniform and steady-flow, whereas the wind flow at the bridge site may be nonuniform in the spanwise direction and unsteady during a VIV event; (3) the Reynolds number of a section model is

generally one to two orders of magnitude smaller than that of a full-scale bridge.

In recent years, structural health monitoring systems (including wind and wind effects monitoring systems) have been implemented for many long-span bridges worldwide (Ko and Ni, 2005; Li et al., 2006; Azarbayejani et al., 2009; Jang et al., 2010; Ou and Li, 2010). These systems collect a large amount of data and provide an opportunity to investigate the wind effects for a prototype bridge based on field full-scale measurements. However, the wind and wind effects for prototype bridges are much more complicated than those in wind tunnel experiments and numerical simulations. Additional challenges are created by the uncertainty associated with bridges and measurement noise. Fortunately, machine learning (ML), which is a branch of artificial intelligence (AI), and which belongs to data-driven methodology, is a powerful tool for investigating wind and wind effects in a complicated measured dataset. ML originated in the 1940s, and in recent years has attracted more attention because of the emergence and development of artificial intelligence. To characterize the flow around a cylinder from limited pressure measurements, Bright et al. (2013) used a supervised ML strategy to compute library elements, encoding the dimensionally reduced dynamics for various Reynolds numbers. Convex L_1 optimization was then used with a limited number of pressure measurements on the cylinder to reconstruct (or decode) the full pressure field and resulting flow field around the cylinder. In terms of turbulence modeling, Zhang and Duraisamy (2015) investigated the ability of ML models to reconstruct the functional forms of spatially distributed quantities extracted from high-fidelity simulation and experimental data. The results highlight the potential of ML as an enabling tool for data-driven turbulence modeling. In addition, Wu and Kareem (2011) developed an approach to model aerodynamic nonlinearities in the time domain utilizing an artificial neural network (ANN) framework with an embedded cellular automata (CA) scheme based on wind tunnel experiments with section models. The applications of ML in wind engineering are currently very naïve, and more attention to this topic is required.



Fig. 1. Location of the investigated bridge.

As mentioned above, structural health monitoring systems have collected significant longitudinal data on wind and the wind-induced vibrations of bridges. In this measured data, VIV responses have been observed for some long-span bridges. This study proposes a data-driven approach with an ML scheme for modeling and predicting the VIV response of a bridge, based on a database of field full-scale measured VIVs for a long-span suspension bridge over six recent years (from 2010 to 2015). First, a decision tree learning algorithm is applied to train the VIV mode classifier, which can identify (predict) the VIV mode from wind speeds and directions. Second, the support vector regression (SVR) algorithm is employed to separately model the VIV response in the time domain for each measured VIV mode. Third, a parametric analysis of VIV response for the bridge is carried out using the trained SVR model, and new findings are obtained from the data-driven model.

2. Full-scale measurements

The long-span suspension bridge investigated in this study has a 1650 m central main span and a 578 m side span. The bridge, which is located in the eastern ocean in China, crosses a narrow water channel that lies between two islands, as shown in Fig. 1. The system that monitors wind speed and vibrations for the bridge was implemented in 2009. It is noted that the existence of a hill on the south island close to the 3/4 center span location leads to the nonuniform flow of natural wind (Li et al., 2014a).

Wind speed at the site is monitored with Young Model 81000 three-dimensional ultrasonic anemometers at a sampling frequency of 32 Hz. The vertical acceleration of the bridge deck is monitored by GT02 force-balance triaxial accelerometers at a sampling frequency of either 100 Hz (in 2010 and 2011) or 50 Hz (from 2012 to 2015; the sampling frequency was modified to reduce the amount of collected data). The anemometers and accelerometers are installed on both the upriver and downriver sides of the sections at the 1/4, 1/2 and 3/4 center span, as shown in Fig. 2. Note that the anemometers are installed on lighting columns at a height 6 m above the bridge deck surface. All of the wind data analyzed in this study are obtained from the inflow anemometers, which can measure natural winds without interference from bridge components. 90° and 270° are defined as the wind directions perpendicular to the bridge axis.

Vertical VIV events for six different modes (frequencies) have been observed on this bridge. The measured time history of vertical acceleration (a-i, b-i, c-i, d-i, e-i, f-i in Fig. 3), the power spectral density (PSD) of the measured acceleration (a-ii, b-ii, c-ii, d-ii, e-ii, f-ii in Fig. 3), and the corresponding measured 10-min mean wind speed history (a-iii, b-iii, c-iii, d-iii, e-iii, and f-iii in Fig. 3) are shown in Fig. 3. To identify the modes of the observed VIV, the frequencies and mode shapes of the bridge are calculated by a finite element method (FEM, the more detailed information of FEM can be found in Jin's master thesis, 2011) and shown in Fig. 3 (a-iv, b-iv, c-iv, d-iv, e-iv, f-iv). It can be seen from Fig. 3 that a complete VIV event may last for hours, and includes a developing stage,

steady stage, and vanishing stage. To de-noise the measured data, statistics over a period are applied for modeling in this study rather than instantaneous measured data. An appropriate time interval for statistics should be determined. If the time interval for statistics is too long, detailed information on VIV response and wind conditions may be lost. Conversely, if the time interval is too short (for example, the sampling interval is selected as the time interval), the impact of noise may be too significant, resulting in the failure of the data-driven modeling of the VIV response. As a compromise, an 1-min interval is selected as the time interval for statistics. Each VIV event is accordingly represented as a time series consisting of 1-min samples.

3. Identification of VIV mode by the decision tree learning algorithm

VIVs on a long-span bridge may occur for different single modes. The damping ratios of different modes may be very different. Structural damping is a critical and sensitive parameter for VIV (Chen, 2013). However, the damping ratio is very scattered, so it is difficult to accurately identify structural damping for a prototype long-span bridge. To minimize the effect of the inaccurate identification of damping ratio, a model is established in this study for predicting the VIV response for each measured VIV mode. Modal damping is accordingly assumed to be constant in the modeling for the VIV mode, and does not need to be included in the input parameters. This modeling approach requires the identification of the VIV mode, by using the wind characteristics in advance to determine which modal VIV response model will be then used for the prediction.

A long-span bridge has a number of modes. The VIV mode (frequency) can be determined from the wind speed and direction according to the Strouhal law and “lock-in” phenomenon, as shown in Fig. 4, under the assumption that the oncoming wind flow is uniform and that wind direction is constant and perpendicular to the bridge axis. In Fig. 4, f_{s1} and f_{s2} are the first and second order natural frequencies of the structure, respectively. However, the actual wind flow for the prototype bridge is much more complex: the oncoming wind flow may be nonuniform in the spanwise direction due to local geomorphic features, the wind direction may not be constant, and the wind flow may be unsteady during a VIV event. The identification or prediction of the VIV mode from wind characteristics for the prototype bridge thus becomes difficult.

An ML algorithm is a powerful tool for classification. Because a few VIV modes occur on the studied bridge, the identification of VIV mode is a multiple-classification problem. The decision tree learning algorithm is capable of handling this type of problem.

3.1. Decision tree learning algorithm

A decision tree learning algorithm is employed in this study to establish a classification model that predicts the VIV mode using the wind characteristics only. The algorithm starts from a root node, and then many child nodes gradually grow, forming a tree structure. A simple example is shown in Fig. 5 to illustrate how a decision can be implemented by a decision tree.

As can be seen in Fig. 5, input parameters should first be selected. The specific attributes (temperature or wind speed) and threshold at each node are determined by the algorithm, which aims to decrease the impurity of training samples in the node by splitting the training samples into two subsets. The impurity of samples in a node (N) is defined in this study by a Gini impurity index:

$$I(N) = \sum_{i \neq j} P(\omega_i)P(\omega_j) = 1 - \sum_j P^2(\omega_j) \quad (2)$$

where $P(\omega_i)$ is the fraction of samples of VIV mode ω_i ($i = 1, 2, 3, 4, 5, 6$) in node N . $I(N)$ reaches a minimum (zero) when the node is pure (all samples in this node belong to VIV mode ω_i ; i.e., only one VIV mode is

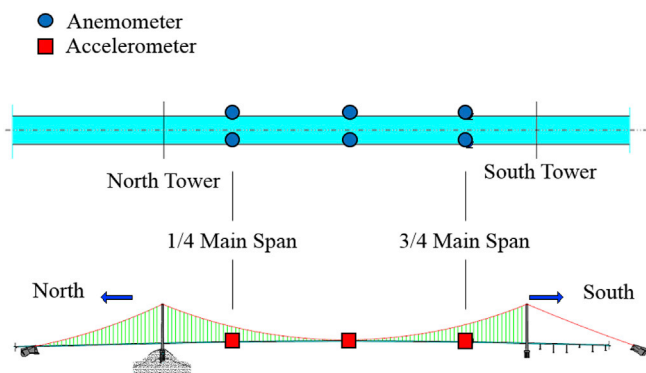


Fig. 2. Locations of anemometers and accelerometers.

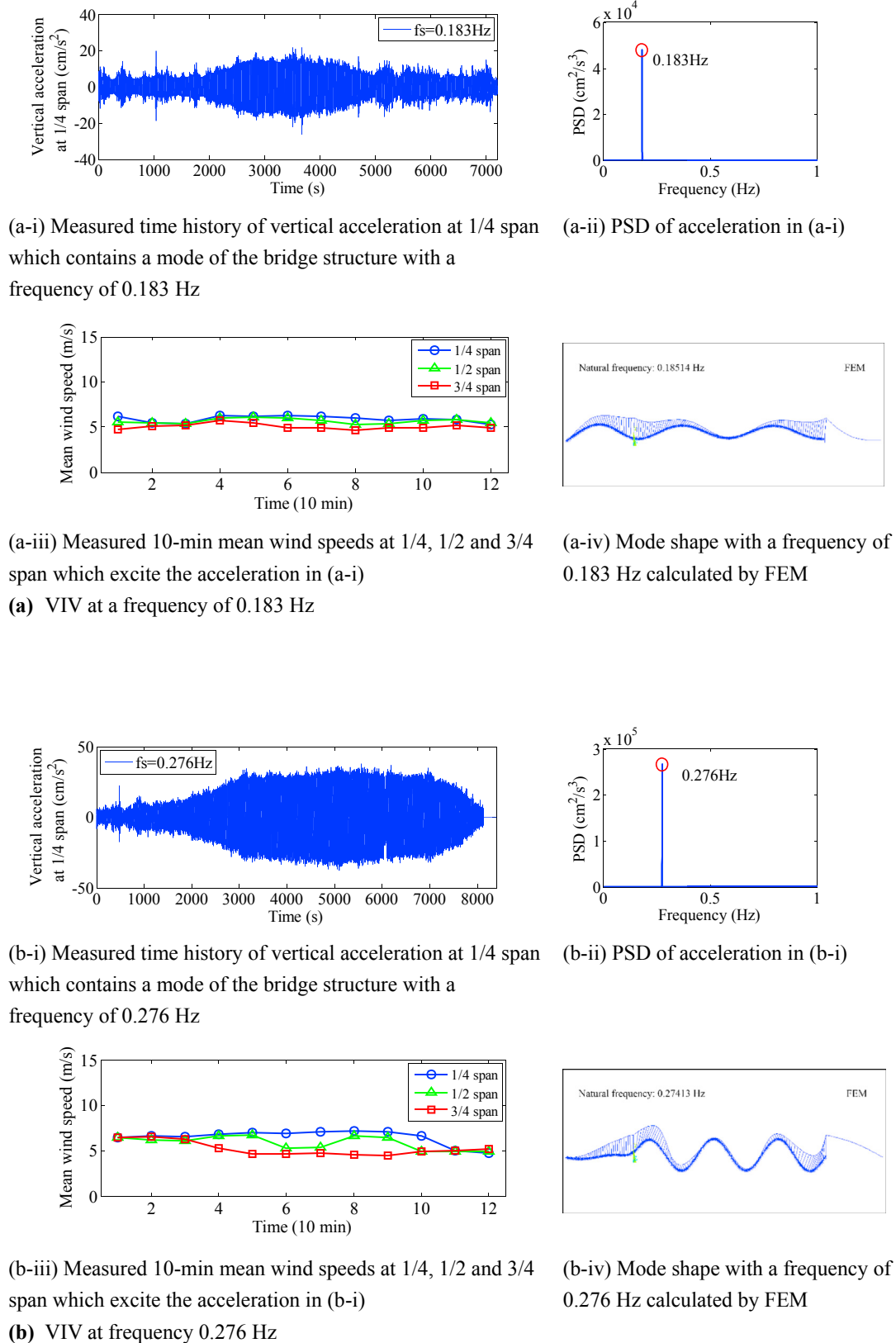


Fig. 3. Measured VIV events of different modes. (a) ~ (f) denote six VIV modes, respectively. In each VIV mode, the top left subfigure is the vertical acceleration history, top right subfigure is the PSD of the acceleration, the corresponding 10-min mean wind speed history is at bottom left, and the FEM-calculated mode shape and natural frequency are at bottom right.

included in the node); and reaches a maximum when the fraction of samples of each VIV mode is the same.

If one node contains only samples of a single VIV mode, complete purity is achieved, with the lowest possible impurity value (zero). This

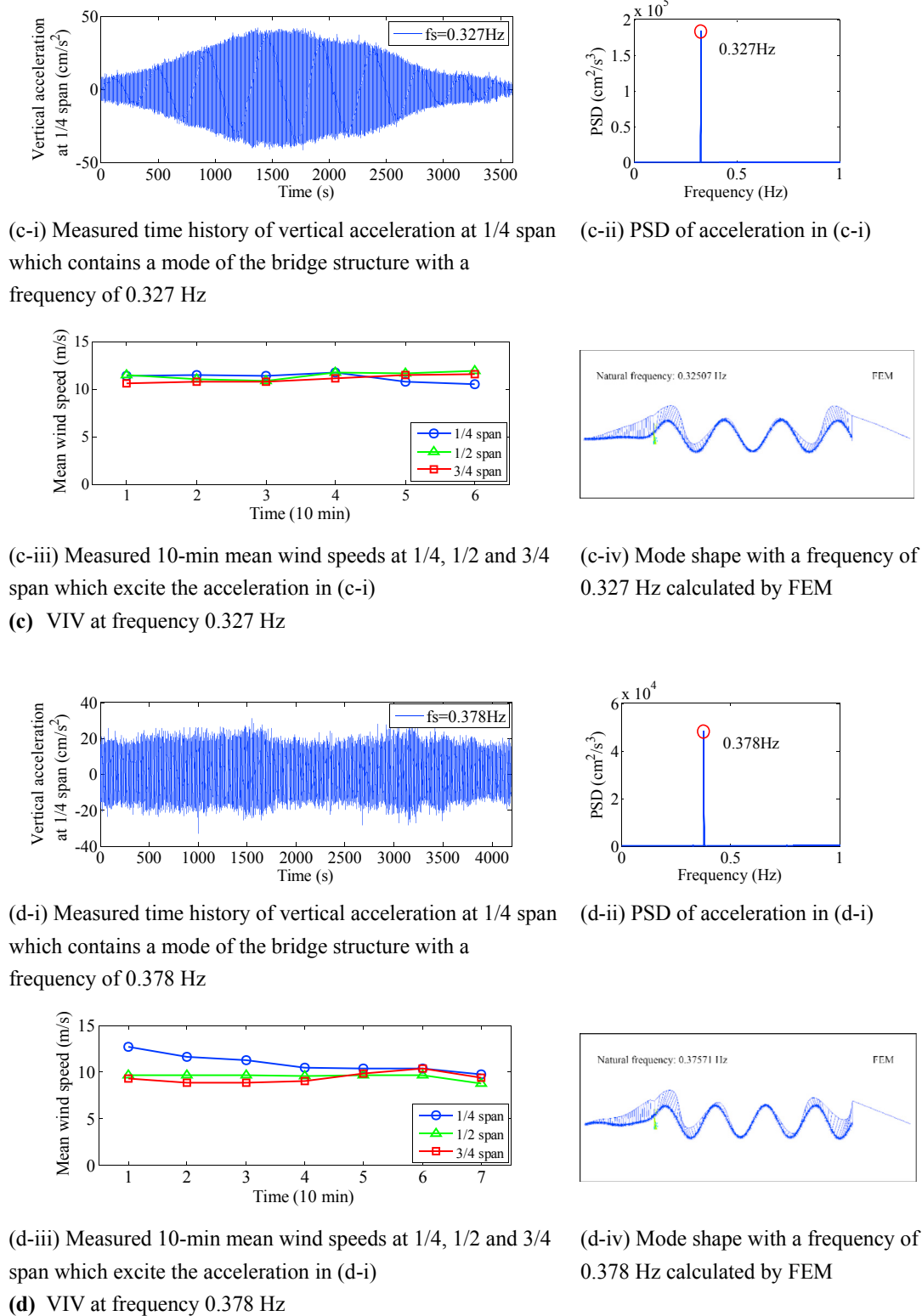
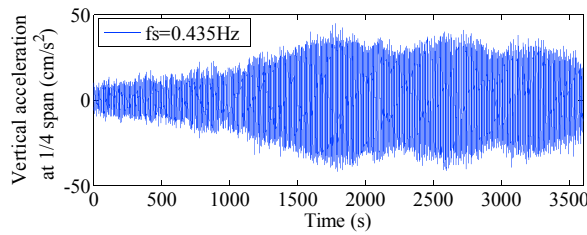
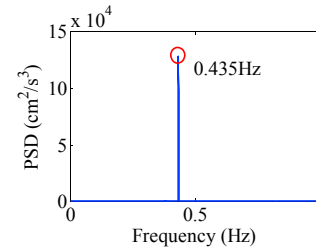


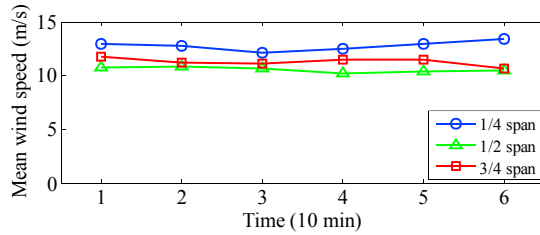
Fig. 3 (continued).



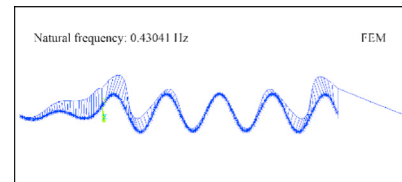
(e-i) Measured time history of vertical acceleration at 1/4 span which contains a mode of the bridge structure with a frequency of 0.435 Hz



(e-ii) PSD of acceleration in (e-i)

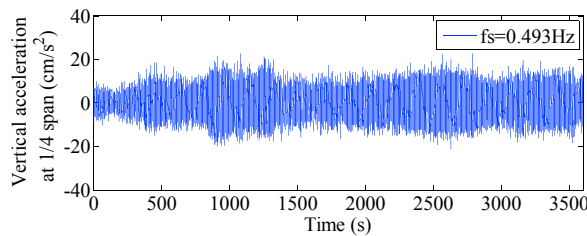


(e-iii) Measured 10-min mean wind speeds at 1/4, 1/2 and 3/4 span which excite the acceleration in (e-i)

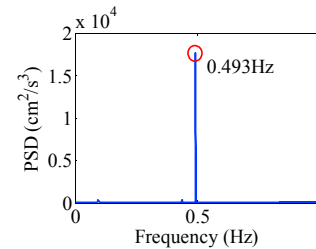


(e-iv) Mode shape with a frequency of 0.435 Hz calculated by FEM

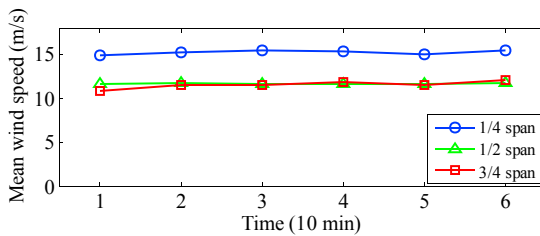
(e) VIV at frequency 0.435 Hz



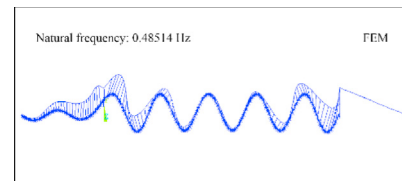
(f-i) Measured time history of vertical acceleration at 1/4 span which contains a mode of the bridge structure with a frequency of 0.493 Hz



(f-ii) PSD of acceleration in (f-i)



(f-iii) Measured 10-min mean wind speeds at 1/4, 1/2 and 3/4 span which excite the acceleration in (f-i)



(f-iv) Mode shape with a frequency of 0.493 Hz calculated by FEM

(f) VIV at frequency 0.493 Hz

Fig. 3 (continued).

node can be labeled with the VIV mode. Therefore, we should determine the threshold and attributes among input parameters that decrease impurity as much as possible, by splitting the training samples in the node into two subsets. The objective function for each splitting at a node is

as follows:

$$\max_{\text{attribute, threshold}} \Delta I(N) = I(N) - [\alpha_1 I(N_1) + \alpha_2 I(N_2)] \quad (3)$$

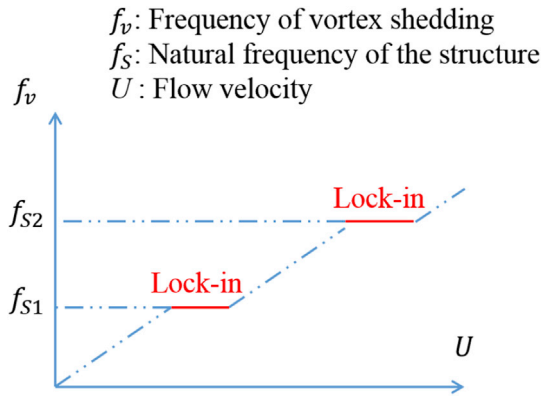


Fig. 4. Vortex-shedding frequency inside and outside “lock-in”. The vortex-shedding frequency keeps constant when the flow velocity is in the “lock-in” range corresponding to a natural frequency of the structure.

where α_1 and α_2 are the fractions of training samples sent down to the child nodes N_1 and N_2 from the parent node N , respectively, and $\alpha_1 + \alpha_2 = 1$; $I(N_1)$ and $I(N_2)$ are the Gini impurity values at nodes N_1 and N_2 , respectively; $\Delta I(N)$ is the weighted decrease in Gini impurity from node N down to nodes N_1 and N_2 .

To find the optimal decision (i.e., determine the attribute and threshold at a node), an extensive, exhaustive search over all samples in the node must be performed. The attribute is determined from all input parameters, and the corresponding threshold is determined from all of the medians between adjacent samples for the parameter in node N . The optimal attribute and its threshold will produce a maximum decrease in Gini impurity.

Beginning with the root node, this process is recursively repeated at each node for the remaining samples, leading to the growth of the decision tree. The recursion is stopped when the samples at a node belong to a single VIV mode.

However, if the tree grows fully until each leaf node achieves the lowest impurity (zero), the model is likely to be overfit. In extreme but rare cases, each leaf node contains only one sample, meaning that each leaf node has a single VIV mode. The tree is just a convenient implementation of a lookup table, and thus cannot be expected to have generalized performance. In this study, the “MinLeaf,” which is the minimum number of samples in the leaf node, is constrained and optimized by a cross-validation method to control the size of the tree and suppress the extent of overfit.

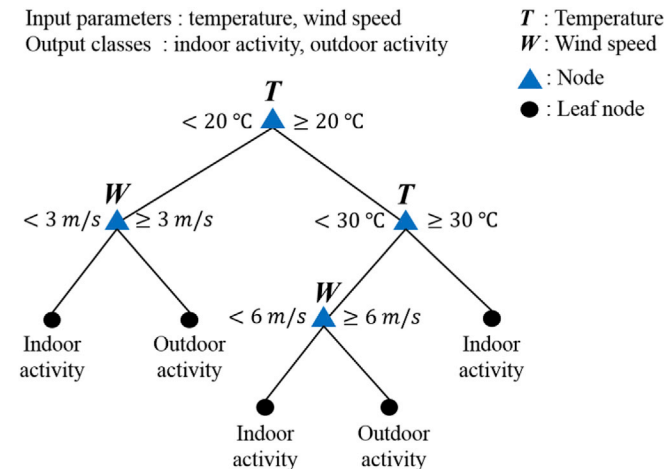
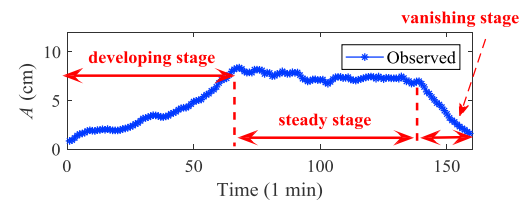


Fig. 5. A simple decision tree example: decide whether to do indoor activity or outdoor activity depending on the weather. From the top node, a series of sub-decisions are made according to the attributes and thresholds on nodes along branches of the tree, and a final decision comes at a leaf node.

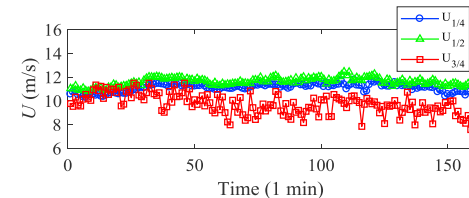
Before conducting cross-validation, the entire sample set is divided into a training set and a testing set. In the cross-validation for searching the optimal “MinLeaf,” the training set is randomly partitioned into m identically sized subsets. One subset is used for validation, and the other $m-1$ subsets are employed for training the decision tree model. The validation subset is selected in turns over all m subsets. In this way, the training is repeated m times, with each of the m subsets used only once for validation. The prediction accuracy is then averaged to produce an estimate given a specified “MinLeaf.” A series of “MinLeaf” values are selected, and the above procedure is repeated. Then the optimal “MinLeaf” is obtained, which corresponds to the model with the best prediction accuracy. With the optimal “MinLeaf” obtained by cross-validation, all training samples are used for training the decision tree model (obtaining the attributes and threshold at all nodes). Finally, the performance of the decision tree model is validated using the samples in the testing set (note that the samples in the testing set never participate in the training of the decision tree model).

3.2. Identification of VIV mode by decision tree learning algorithm

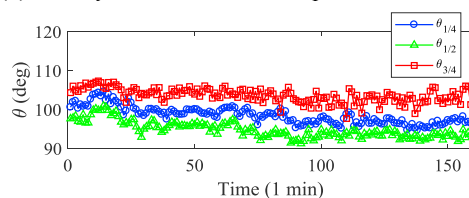
In this study, the displacement response at the 1/4 center span location is first calculated by integrating the measured the vertical acceleration in the frequency domain. The mean wind speed (U), mean wind direction (θ), and RMS of the vertical displacement at 1/4 center span (A) are calculated for each 1-min sample. Each VIV event can thus be represented as a time series of vibration displacement amplitudes (1-min RMS) and the corresponding time series of 1-min wind characteristics, as shown in Fig. 6. It can be observed that the VIV event includes a developing stage (amplitude increase), steady stage (near-constant amplitude), and vanishing stage (amplitude decrease). The wind is a temporal and spatial variable during the VIV event. The wind condition in the developing stage is the most important in the formation of the VIV. The wind characteristics in this stage must be relatively steady, and must meet the necessary conditions under which vortices periodically shed at a frequency approaching a certain structural frequency. In the steady stage, the wind effects of the aeroelastic system are primarily controlled by large-amplitude vibration. The required wind conditions thus become



(a) History of 1-min RMS of vertical displacement at 1/4 center span



(b) History of 1-min mean wind speed



(c) History of 1-min mean wind direction

Fig. 6. The RMS of vertical displacement and corresponding wind flow in a VIV event.

less rigorous, and the wind can vary within a certain range. In the vanishing stage, the wind velocity exceeds the “lock-in” range or becomes very unsteady. After explanation of the three stages, it becomes easy to understand that the mode (frequency) of a VIV event is mainly determined by the oncoming wind during the formation of the VIV in the developing stage. Therefore, all 1-min samples in the developing stage of each VIV event are extracted to establish a dataset to train the model for the identification of a VIV mode based on wind characteristics. The developing stage is quantitatively defined as the duration from the beginning of the identified VIV event to the maximal vibration amplitude.

To classify the VIV mode, the input parameters should be first determined. Because the wind is nonuniform along the bridge axis, and the wind direction may change, it is better to represent the wind field by using the monitored wind at as many locations as possible along the bridge span. In the investigated bridge, six anemometers have been installed at three locations along the span, as shown in Fig. 2 (upriver and downriver of the 1/4, 1/2 and 3/4 center span). Therefore, for establishing a predictive model of the VIV mode, six parameters including $[U_{1/4} \ U_{1/2} \ U_{3/4} \ \theta_{1/4} \ \theta_{1/2} \ \theta_{3/4}]$ ($U_{1/4}$, $U_{1/2}$, $U_{3/4}$ and $\theta_{1/4}$, $\theta_{1/2}$, $\theta_{3/4}$ are the horizontal mean wind speeds and mean wind directions of the inflow wind at 1-min intervals at the 1/4, 1/2, and 3/4 center span locations, respectively) are selected as the input parameters. The VIV mode represented by the predominant frequency of the measured vertical vibration at the 1/4 center span is the class label. For a clear observation, a total of 2275 samples are depicted in a three-dimensional space, with horizontal mean wind speeds at the locations of the 1/4, 1/2, and 3/4 center spans (see Fig. 7).

The 2275 samples are randomly divided into a training set (2000 samples) and a testing set (275 samples). For conducting the cross-validation, the 2000 training samples are randomly divided into ten subsets (each subset contains 200 samples). One of the ten subsets is selected for validation in turns over all ten subsets, while the nine remaining subsets are used to train the decision tree structure using the algorithm in section 3.1 for obtaining the optimal “MinLeaf.” Based on the optimal “MinLeaf,” the decision tree model is re-trained using all 2000 samples in the training set. The attributes and threshold at all nodes are then determined through the training. The 275 samples in the testing set are employed to validate the performance of the decision tree model.

3.3. Results of identification of VIV modes

To obtain the optimal “MinLeaf,” ten-fold cross-validation is carried out on the training set. The results are shown in Fig. 8(a). It can be seen from Fig. 8(a) that the optimal “MinLeaf” is four, and that the corresponding maximum averaged prediction accuracy for the cross-

validation is 93.55%. Thus, the decision tree is trained using all samples in training set with a “MinLeaf” of four to obtain the attributes and threshold of all nodes. The overall decision tree model is shown in Fig. 9 where M1, M2, M3, M4, M5, and M6 indicate the modes of 0.183 Hz, 0.276 Hz, 0.327 Hz, 0.378 Hz, 0.435 Hz, and 0.495 Hz, respectively. The decision tree model is then tested on the testing set, and the results are shown in Fig. 8(b). The testing accuracy is 94.18%, indicating that 94.18% of the 1-min testing samples are correctly classified.

4. Modeling of VIV response using SVR

Once various VIV modes are identified, the modeling of VIV response is further studied. Because of the abovementioned uncertainty associated with damping ratio, a VIV response model should be established for each VIV mode. As mentioned in section 3, the wind conditions for a prototype bridge are very complicated. ML algorithms are much better suited for modeling such complex problems. An SVR algorithm is employed to establish a VIV response model. SVR algorithms have good generalization capacity and can use the kernel method to overcome the curse of dimensionality, referring to various phenomena that arise when analyzing data in high-dimensional spaces but do not occur in low-dimensional spaces.

Each VIV event is represented by a time series of vibration displacement amplitudes (1-min RMS) at the 1/4 center span location, and the corresponding time series of wind characteristics. The wind is the input and the vibration response is the output for the VIV of the dynamic system. For a time series of 1-min responses produced in the dynamic system, the vibration amplitude response at the n th step can be determined by the wind characteristics at the same moment and the vibration amplitude response at the $n-1$ st step, which can be described as follows

$$A^n = f_0 \left(A^{n-1}, U_{1/4}^n, U_{1/2}^n, U_{3/4}^n, \theta_{1/4}^n, \theta_{1/2}^n, \theta_{3/4}^n \right) \quad (4)$$

where $f_0(\bullet)$ is the unknown function (model) that reflects the characteristics of the dynamic system; A is the 1-min RMS of the displacement at the 1/4 center span location; the superscript indicates the time step; A^{n-1} is vibration amplitude at previous step. The specific objective of the modeling of the VIV response is to approximate the unknown function f_0 for a large number of measured data samples. Once the model is established, the VIV response can be predicted in a recursive process, given an initial vibration amplitude f_0 and real-time wind condition $\{(U_{1/4}^n, U_{1/2}^n, U_{3/4}^n, \theta_{1/4}^n, \theta_{1/2}^n, \theta_{3/4}^n)\}_{n=1,2,3,\dots}$ with the vibration amplitude predicted at the current time step being used as a part of the input at next time step.

4.1. SVR algorithm

The objective of SVR is to approximate an unknown function $g(x)$ from a given set of samples $\{(x_i, y_i)\}_{i=1}^M$ that are obtained from the function $g(x)$, where each x_i denotes the sample inputs and has a corresponding target value y_i , and M corresponds to the size of the training set. The basic concept is to affine the training data from the input space into a higher dimensional feature space via nonlinear mapping (Φ), and to perform a linear regression in this feature space (Cortes and Vapnik, 1995; Smola and Schölkopf, 2004). In this study, the input x is the vector that consists of the VIV response at the $n-1$ st step and the wind conditions at the n th step, which are represented by $(A^{n-1}, U_{1/4}^n, U_{1/2}^n, U_{3/4}^n, \theta_{1/4}^n, \theta_{1/2}^n, \theta_{3/4}^n)$; and the target variable y is the VIV response at n th step, represented by A^n . The regression function $f(x)$ can be written as

$$f(x) = \langle w, \Phi(x) \rangle + b \quad (5)$$

where $w \in R^d$ and $b \in R^d$, d is the dimension of the feature space; $\langle \cdot, \cdot \rangle$ represents the dot product; and Φ is the nonlinear mapping, which can be

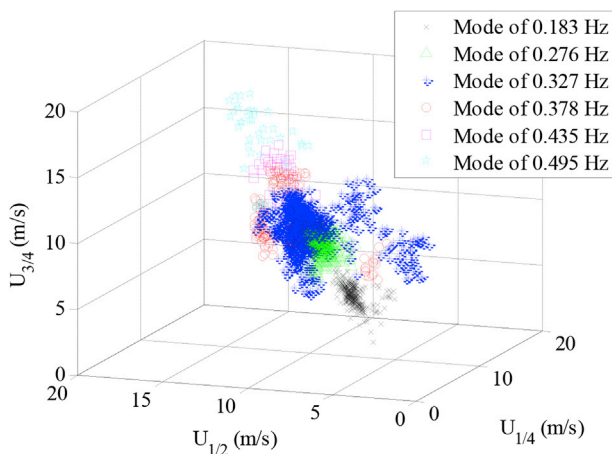
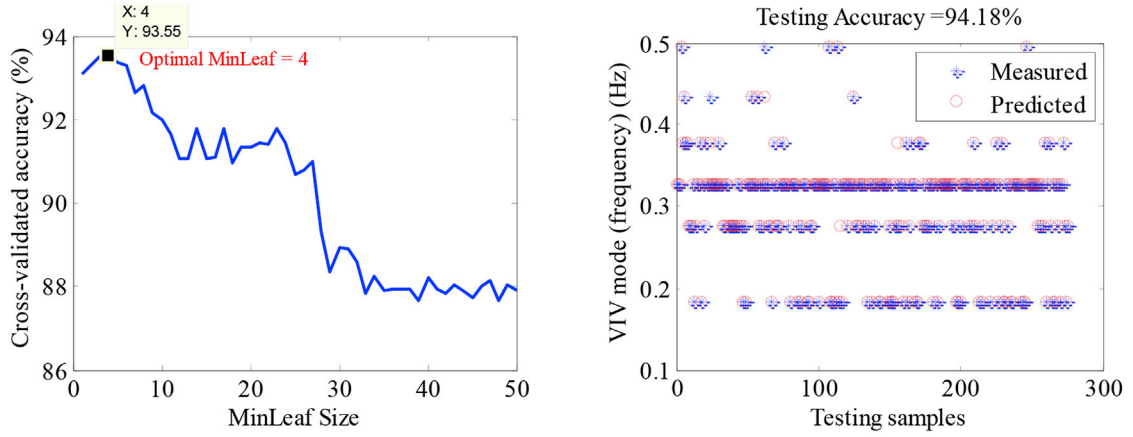


Fig. 7. Wind speed samples represented as points in a three-dimensional wind speed space for visualization (wind direction information is not given for the observation).



(a) Accuracy achieved in cross-validation

(b) Prediction of VIV frequency on testing set

Fig. 8. Results of the decision tree learning: (a) cross-validation result of optimizing the parameter MinLeaf; and (b) prediction of VIV frequency on testing set by the model learned with the obtained optimal MinLeaf in (a).

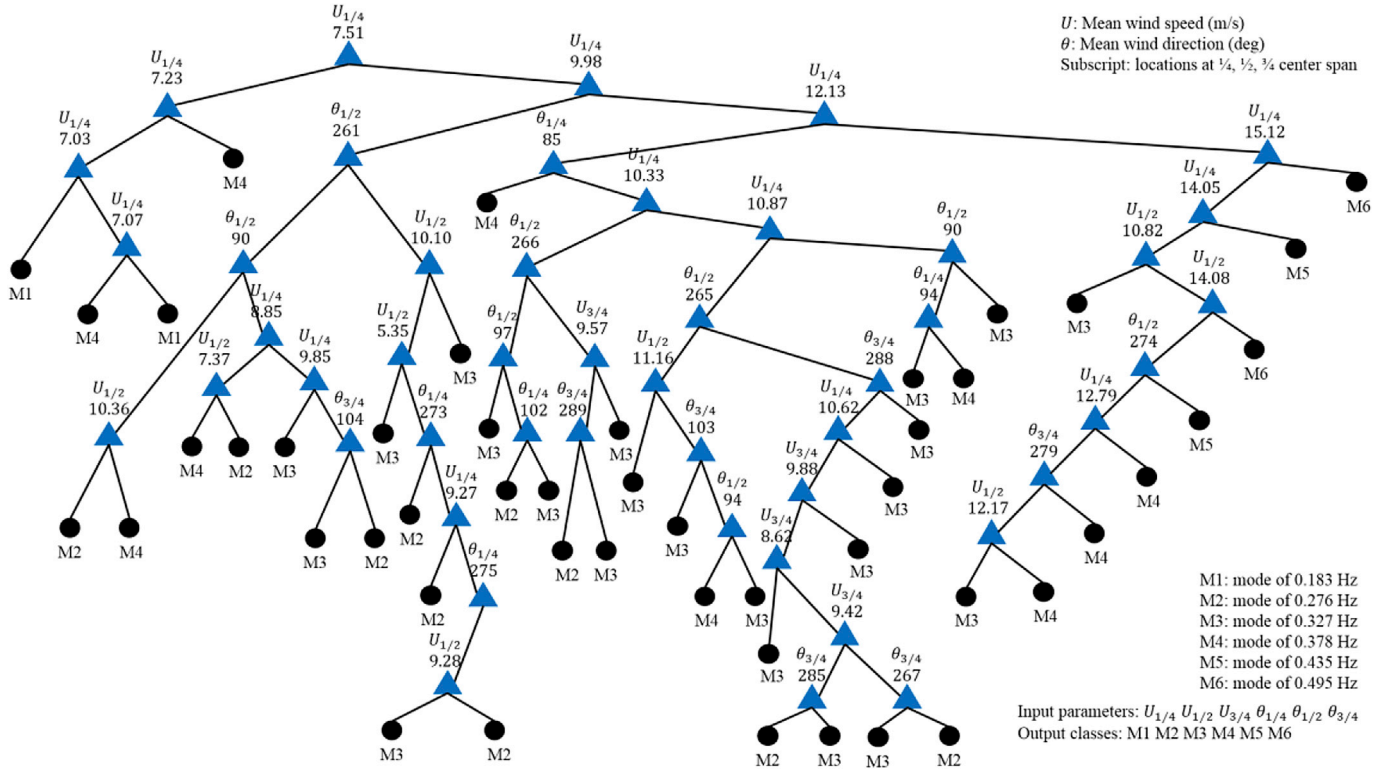


Fig. 9. Overall structure of the obtained decision tree model for identifying VIV mode of the bridge (input parameters: $U_{1/4}$ $U_{1/2}$ $U_{3/4}$ $\theta_{1/4}$ $\theta_{1/2}$ $\theta_{3/4}$; output classes: M1 M2 M3 M4 M5 M6).

described in a kernel function defined in this study as a Gaussian radial basis function:

$$K(x, x') = \langle \Phi(x), \Phi(x') \rangle = \exp(-\gamma \|x - x'\|^2) \quad (6)$$

where $K(x, x')$ is Gaussian radial basis kernel function, γ is a parameter determining the radial range that must be prescribed, and x' is an additional input. In Eq. (5), the coefficient vector w determines the orientation of a discriminating hyperplane, whereas b determines the offset of the discriminating hyperplane from the origin. We aim to find the values of w and b that minimize the following function with an ε - intensive loss function

$$\begin{aligned} \min \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^M (\xi_i + \xi_i^*) \\ \text{s. t.} \quad & \begin{cases} y_i - \langle w, \Phi(x_i) \rangle - b \leq \varepsilon + \xi_i \\ \langle w, \Phi(x_i) \rangle + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \end{aligned} \quad (7)$$

where $\varepsilon \geq 0$ and is a prescribed tolerance. The deviation for the obtained regression function is no larger than ε for any given training data. Every training sample has the corresponding slack variables ξ_i and ξ_i^* , which are used to determine whether the training sample is in or out of the range of

ε . C is a penalty parameter, which weights the tradeoff between the fitting errors and the flatness of f , and which must be prescribed.

To facilitate the solution of Eq. (7), the problem is transformed to a dual space representation using positive Lagrangian multipliers $\alpha_i, \alpha_i^*, \beta_i, \beta_i^*$ as

$$L = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^M (\xi_i + \xi_i^*) - \sum_{i=1}^M (\beta_i \xi_i + \beta_i^* \xi_i^*) - \sum_{i=1}^M \alpha_i (\varepsilon + \xi_i - y_i + \langle w, \Phi(x_i) \rangle + b) - \sum_{i=1}^M \alpha_i^* (\varepsilon + \xi_i^* - y_i + \langle w, \Phi(x_i) \rangle + b) \quad (8)$$

where L is the Lagrangian function. Solving the Lagrangian dual problem, a pair of Lagrangian multipliers (α_i, α_i^*) will be obtained for each training sample. Additionally, the coefficient vector w can be obtained according to the Karush-Kuhn-Tucker (KKT) conditions (Karush, 1939; Kuhn and Tucker, 1951) as follows:

$$w = \sum_{i=1}^M (\alpha_i - \alpha_i^*) \Phi(x_i) \quad (9)$$

Substituting Eqs. (6) and (9) into Eq. (5), the regression function can be rewritten with Lagrangian multipliers as

$$f(x) = \sum_{i=1}^M (\alpha_i^* - \alpha_i) \exp(-\gamma \|x_i - x\|^2) + b \quad (10)$$

It can be seen that the regression function is completely described as a combination of the training samples x_i . It is noted that the values of both α_i and α_i^* for a majority of training samples are zero, and only samples with nonzero α_i and α_i^* values (the so-called support vectors) contribute to regression function. The value of b can be obtained by any one of the support vectors according to the Karush-Kuhn-Tucker (KKT) conditions (Karush, 1939; Kuhn and Tucker, 1951)

$$\begin{aligned} b &= y_i \langle w, \Phi(x_i) \rangle - \varepsilon \quad \text{for } \alpha_i \in (0, C) \\ b &= y_i \langle w, \Phi(x_i) \rangle + \varepsilon \quad \text{for } \alpha_i^* \in (0, C) \end{aligned} \quad (11)$$

In this paper, the three parameters (C, γ, ε) to be prescribed are determined by an optimization based on a combination of cross-validation and genetic algorithms (Whitley, 1994). The SVR algorithm is implemented based on the library for support vector machines (LIBSVM) proposed by Chang and Lin (2011).

4.2. Modeling of VIV response

The database used in this study consists of 73 VIV events with six different modes, as shown in Table 1. The VIVs of high-order modes are less observed in the field due to their exciting conditions of high wind speed. Thus far, the number of VIV events of high-order modes (0.378 Hz, 0.435 Hz, and 0.495 Hz) is too small to ensure the good generalization capacity of the model obtained using ML. Therefore, this study trains the models only for the low-order modes (0.183 Hz, 0.276 Hz, and 0.327 Hz),

Table 1
Number of VIV events in the database used in this study.

Mode (frequency)	Number of VIV events	Number of instances (1-min samples)
0.183 Hz	14	766
0.276 Hz	15	970
0.327 Hz	34	2406
0.378 Hz	6	274
0.435 Hz	1	39
0.495 Hz	3	97

in order to validate the proposed data-driven approach. The modeling for high-order modes can be implemented using the same approach with an accumulation of more VIVs of high-order modes. For testing the models, one VIV event for each mode is randomly selected as the testing set. The Normalized Mean Square Error (NMSE) and Squared Correlation Coefficient (SCC) are defined for measuring the prediction results,

$$\begin{aligned} \text{NMSE} &= \frac{\frac{1}{l} \sum_{i=1}^l (\hat{A}_i - A_i)^2}{\sigma_A^2} \\ \text{SCC} &= \frac{\left(\frac{1}{l} \sum_{i=1}^l \hat{A}_i A_i - \frac{1}{l} \sum_{i=1}^l \hat{A}_i \frac{1}{l} \sum_{i=1}^l A_i \right)^2}{\left(\frac{1}{l} \sum_{i=1}^l \hat{A}_i^2 - \left(\frac{1}{l} \sum_{i=1}^l \hat{A}_i \right)^2 \right) \left(\frac{1}{l} \sum_{i=1}^l A_i^2 - \left(\frac{1}{l} \sum_{i=1}^l A_i \right)^2 \right)} \end{aligned} \quad (12)$$

where l is the number of samples to be predicted; A and $\hat{A}$ are the measured and predicted vibration amplitudes, respectively.

The difference between the training and prediction of the response should be emphasized by two different configurations, as shown in Fig. 10. The series-parallel configuration is used for training (as described in Eq. (4)), while the parallel configuration is used for predicting a VIV response history, which can be described as follows:

$$\hat{A}^n = f(\hat{A}^{n-1}, U_{1/4}^n, U_{1/2}^n, U_{3/4}^n, \theta_{1/4}^n, \theta_{1/2}^n, \theta_{3/4}^n) \quad (13)$$

where f is the function obtained by the SVR, and $\hat{A}$ is the predicted response.

4.3. Modeling results

The optimal parameters (C, γ, ε) for training these models are obtained from a combination of cross-validation and genetic algorithms (Whitley, 1994), and are shown in Table 2. The models for VIV modes of 0.183 Hz, 0.276 Hz, and 0.327 Hz are then obtained based on these parameters. Each of the three models is tested for predictions using one training event and one testing event following the parallel configuration, respectively. Figs. 11–13 shows the prediction results. The obvious nonuniformity of the wind is observed in Fig. 11(a-ii): the measured wind speeds at the 1/4, 1/2 and 3/4 center span locations become very different after approximately 30 min, and while approach to be almost the same after approximately 100 min resulting in a decrease and subsequent increase in vibration amplitude. It is noted that the predicted response shows the identical variation as described above. Therefore, it can be seen that the data-driven model obtained by our proposed method has successfully learned the influence of the temporal and spatial variation of the wind on VIV response. Furthermore, the predicted responses to the testing events are in generally good agreement with the observed responses. However, the prediction of the testing VIV event for the 0.327 Hz mode shows an apparent disagreement with the observed response in the vanishing stage, as shown in Fig. 13(b-i). From the perspective of wind data, the measured wind conditions at the 1/4, 1/2 and 3/4 span locations in the vanishing stage are almost the same as in the steady stage, while the measured response decreases.

The discrepancies between the prediction and observation may be caused by following factors. One of factors is incompleteness of the input data, i.e. wind condition. For instance, wind at three sections only are obtained and used in the data-driven model, while it may be unable to represent the whole wind field around bridge perfectly, especially when the wind shows significant nonuniformity; and wind turbulence is not included in the data-driven model, which may reduce the prediction accuracy with some extent. The second factor may be from the uncertainty of the structural model. As well known, the real bridges suffer from uncertainties for operation conditions. The data-driven model may have

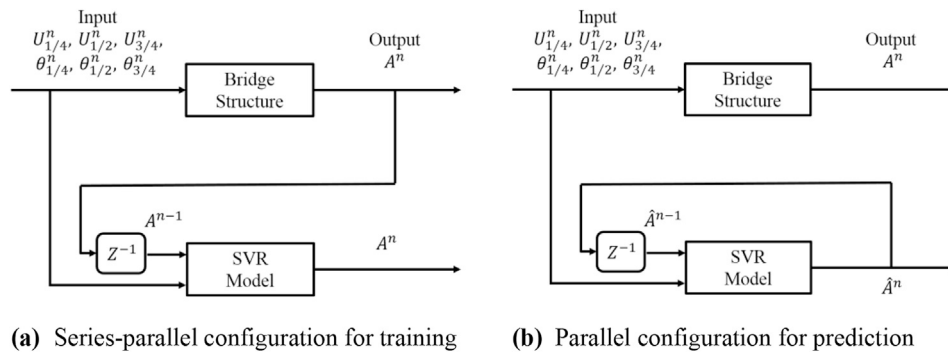


Fig. 10. Configurations for training and prediction. Series-parallel configuration means the vibration displacement amplitude at previous step is from the measurement; while parallel configuration means vibration displacement amplitude at previous step is from the predicted result in a recursive manner.

Table 2

Optimal parameters obtained by a combination of cross-validation and genetic algorithm.

Mode of models	C	γ	ϵ
0.183 Hz	3.308	0.606	0.011
0.276 Hz	4.530	1.188	0.039
0.327 Hz	36.041	0.286	0.020

captured some uncertainty, while loss other uncertainty, i.e. the model may not capture all uncertainty characteristics. The third factor is from the training model. The hyper-parameters (C, γ, ϵ) in these SVR models may take some responsibility for the discrepancies. If the penalty parameter C is too large, we have a high penalty for non-separable points and we may store many support vectors and overfit; while if it is too small, we may have underfitting. The parameter γ in the Gaussian radial

basis kernel function defines how far the influence of a single training sample reaches, with low values meaning ‘far’ and high values meaning ‘close’. If γ is too large, the radius of the area of influence of the support vectors only includes the support vector itself and no amount of regularization with C will be able to prevent overfitting; while if it is too small, the model is too constrained and cannot capture the complexity or “shape” of the data. As for the parameter ϵ , it has an effect on the smoothness of the SVR’s response and it affects the number of support vectors, therefore, both the complexity and the generalization capability of the network depend on its value. An increase in ϵ means a reduction in requirements for the accuracy of approximation. The fourth factor may be attributed to that the number of measured samples is not enough. With increase in samples in the future, the accuracy of the model can be improved.

By integrating the VIV mode classifier (which is established using the

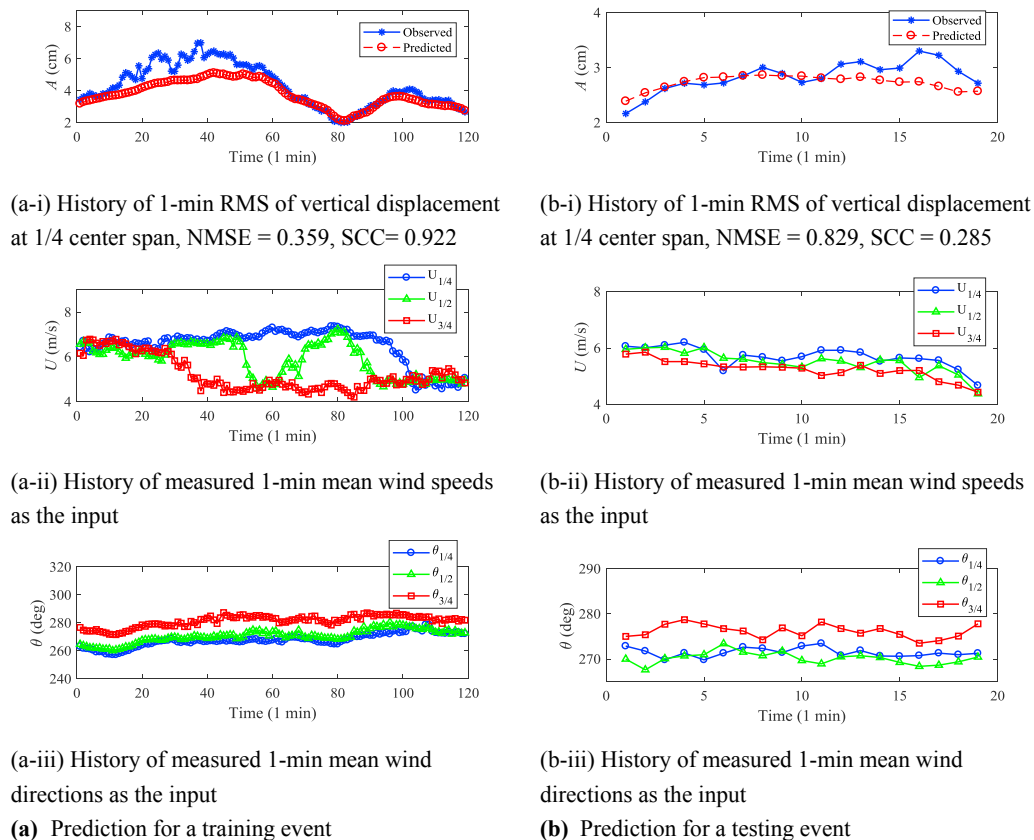


Fig. 11. Predictions of VIV responses for 0.183 Hz mode.

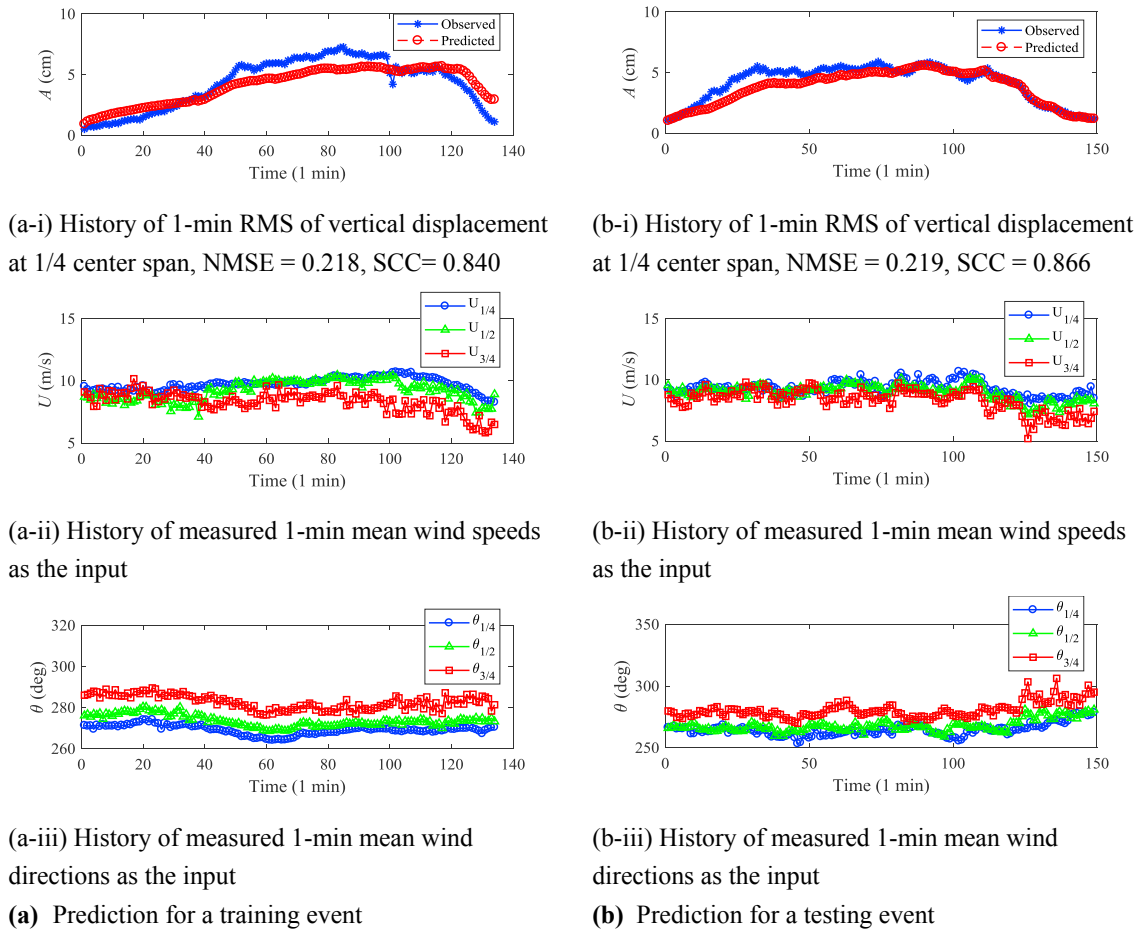


Fig. 12. Predictions of VIV responses for 0.276 Hz mode.

decision tree learning process described in section 3), and creating the SVR-based response prediction model, a complete data-driven approach for VIV modeling and prediction is established. The resulting flowchart is shown in Fig. 14.

5. Parametric analysis of the VIV response

Data-driven models generally lack explicit physical interpretations, and are usually regarded as black boxes that are simply required to affine outputs from inputs. However, the implicit mapping relationship from the inputs and output of a data-driven-based model may contain the physical properties of VIV responses of the bridge, which is important for understanding VIV. To study the physical properties of VIV, a parametric analysis using the trained model is conducted. Based on inputs from the above model, the parameters include initial response, wind speed, wind direction, and wind speed nonuniformity. The parametric analysis is carried out for all three modes, and only the results for the 0.327 Hz mode are presented here (similar results were obtained for the other two modes).

(1) Initial response

Given the assumptions of uniform and steady flow, the inputs include a times series of steady wind characteristics (11, 11, 11, 270, 270, 270), and various initial vibration amplitudes A^0 . The VIV responses for the given inputs are obtained using the trained model, and are shown in Fig. 15. It can be seen from Fig. 15 that the steady-state response is independent from the initial vibration amplitude A^0 . That is, regardless of

the initial state, VIV can be excited by uniform and steady flow for wind speeds and directions within the specified range.

(2) Wind speed and direction

Given the assumption of uniform flow and steady flow, the inputs include the steady wind histories ($U, U, U, 270, 270, 270$) and (11, 11, 11, θ, θ, θ), respectively, and an initial vibration amplitude $A^0 = 0$, which aims to investigate the influence of wind speed and direction on VIV response. The results are shown in Fig. 16. It can be seen from Fig. 16 that the wind speed and direction influence both the steady-state response amplitude and developing duration of VIV from the initial state to the steady state. The steady-state response amplitude and developing duration for each case are extracted and shown in Fig. 17. It can be seen from Fig. 17 (a) that VIV can be excited for wind speeds within an approximate range from 10.2 m/s to 11.8 m/s, with a wind direction of 270° . The phenomenon in which larger steady-state responses correspond to a shorter developing duration times for wind speeds ranging from 10.2 to 11.8 m/s implies more intensive interaction between the vortices and bridge deck. In the majority of previous studies, wind direction is assumed to be perpendicular to the bridge axis. However, in practice, wind direction is time-variant. The results shown in Fig. 17 (b) indicate that the VIV response depends on the wind direction. Specifically, a steady-state response is achieved in a short period of time, and remains approximately constant for the most dramatic VIV responses (those with a wind direction ranging from approximately 260° – 280°); however, once the wind direction is out of the range, the response drops sharply, meaning that the VIV resonance

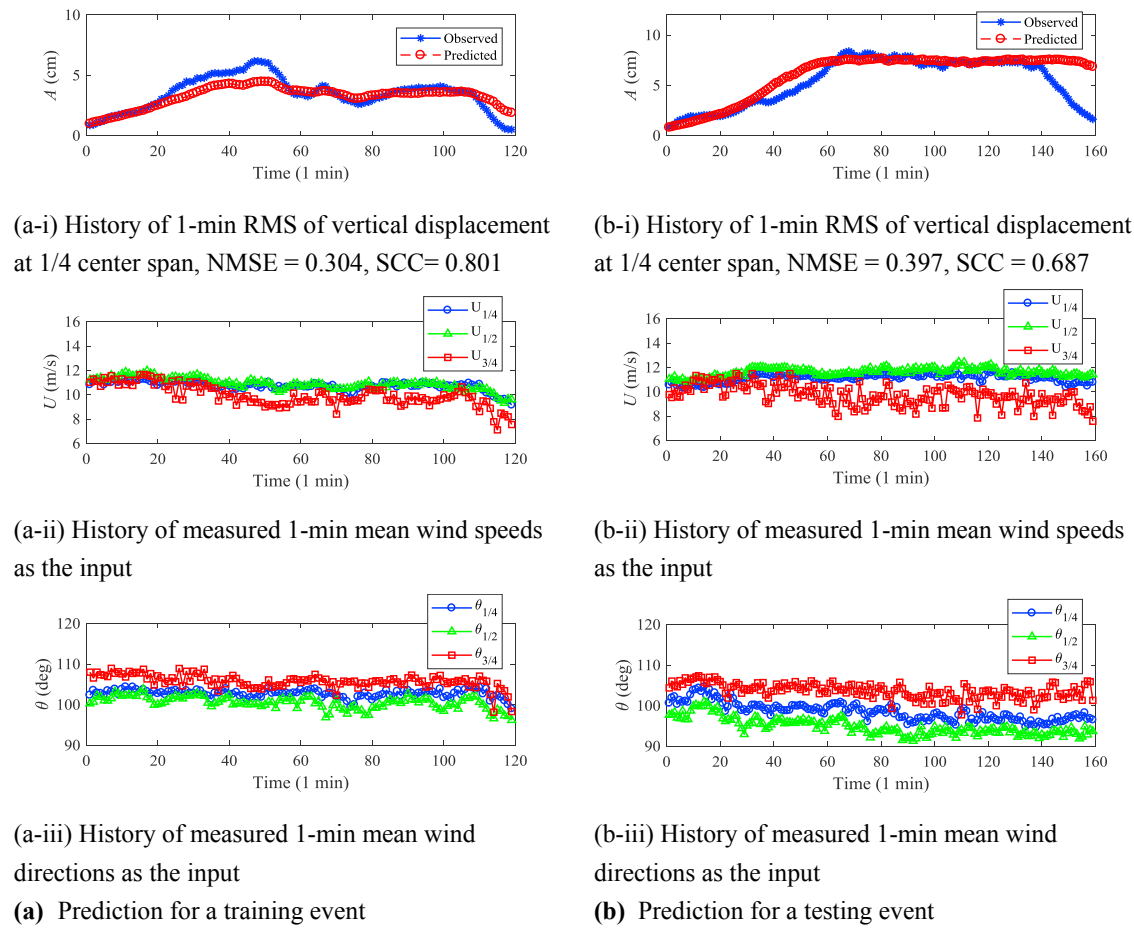


Fig. 13. Predictions of VIV responses for a 0.327 Hz mode.

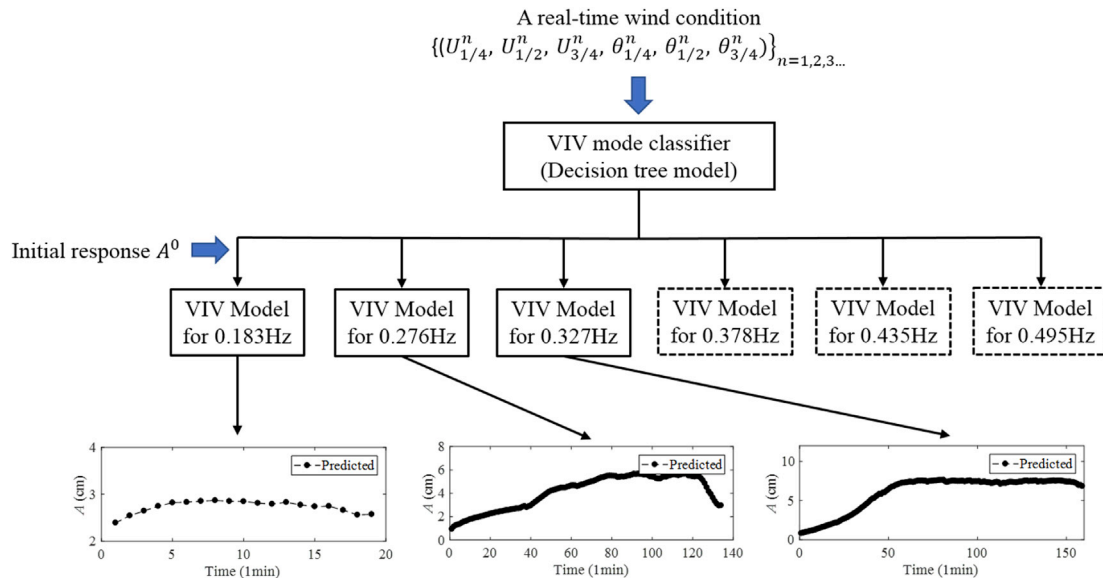


Fig. 14. Flow chart of the prediction of VIV response using the proposed data-driven approach. Given a real-time wind condition, this approach predicts the VIV mode by decision tree model to decide which mode of SVR-based VIV response model is used; and then predict the VIV response by using the corresponding SVR-based model.

vanishes. The developing duration is the shortest for wind directions near 270° (the perpendicular direction to the span), implying that the strongest interactions between the vortices and bridge deck occur for this case. The mechanisms of the effect of wind direction on VIV events need to be studied further.

(3) Nonuniformity of wind speed along bridge span

One of the more interesting findings is that both the steady-state response and developing duration are not symmetric to the center of the “lock-in” wind speed range. This VIV property needs to be further

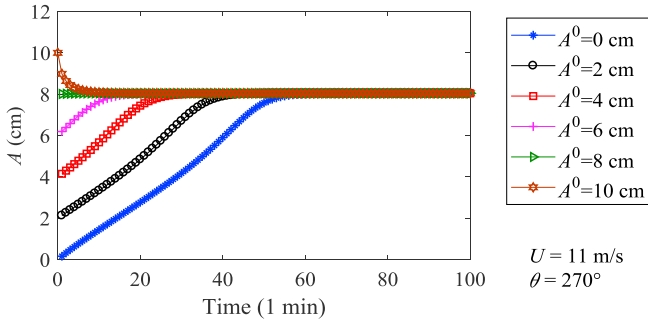
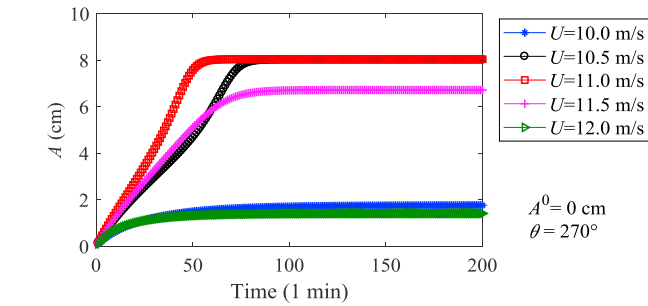


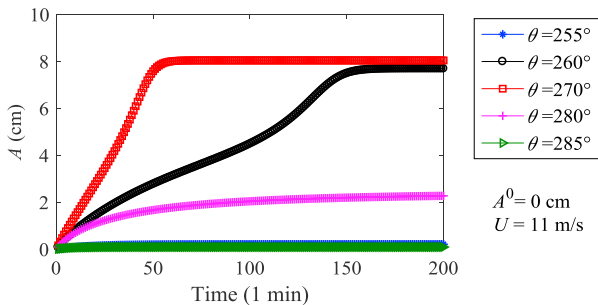
Fig. 15. Influence of initial response on the VIV process.

studied and explained. Another finding is that neither the steady-state response nor developing duration is symmetric to the wind direction (270° , perpendicular to the bridge axis). The asymmetry of the investigated bridge shown in Fig. 2 may account for this phenomenon.

In previous study on VIV in the literature, wind conditions are assumed to be uniform along the bridge span. However, wind speed along the span may be nonuniform for prototype long-span bridges. To investigate the influence of the nonuniformity of wind speed on the VIV, the input of wind speed at three locations is assumed to be $\Delta U = 1.5$, and the initial response A^0 is zero. ΔU represents the level of nonuniformity. $U + \Delta U$, U , and $U - \Delta U$ represent the wind speeds at the 1/4, 1/2 and 3/4 spans, respectively, simulating the real nonuniform wind conditions at the studied bridge site. The steady-state responses under different levels of nonuniformity are obtained from the trained model and are shown in Fig. 18. It can be observed that with the increasing extent of wind speed nonuniformity when ΔU is smaller than 1.5 m/s, the maximum steady-state response changes minimally for a decreasing “lock-in” range of spatially averaged wind speeds $\bar{U} = (U_{1/4} + U_{1/2} + U_{3/4})/3$; however, the steady-state response decreases dramatically when ΔU exceeds 1.5 m/s. This phenomenon may be attributed to the fact that when the level of nonuniformity is lower than $\Delta U = 1.5$ m/s, the wind speeds at all locations along the span fall within the “lock-in” range, and the maximum steady-state response is insensitive to nonuniformity. Increases in ΔU narrow the “lock-in” range of U , because the wind speeds at

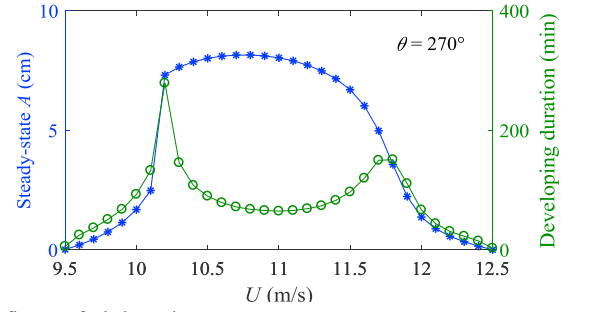


(a) Influence of wind speed

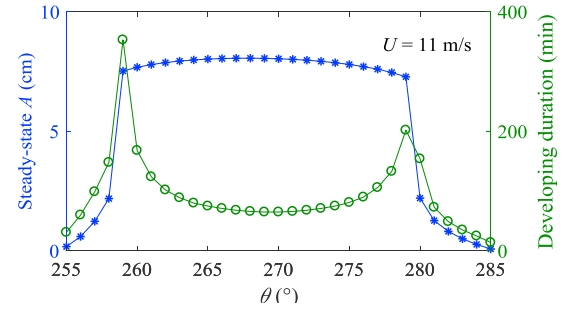


(b) Influence of wind direction

Fig. 16. Influence of wind speed and wind direction on VIV response process.



(a) Influence of wind speed



(b) Influence of wind direction

Fig. 17. Influence of wind speed and wind direction on steady-state response and developing duration.

the 1/4 and 3/4 spans ($U + \Delta U$ and $U - \Delta U$) must also be inside the “lock-in” range. However, when the nonuniformity is above $\Delta U = 1.5$ m/s, wind speeds at some locations along the span are outside the “lock-in” range, and VIV resonance is thus reduced or even eliminated.

6. Conclusions

This paper proposes a data-driven approach with an ML scheme to model the VIV response of a prototype full-scale bridge using a six-year field measurement dataset. The approach includes the identification of VIV mode using a decision tree learning algorithm (i.e., the classification of VIV modes) and the modeling of VIV response for each mode using an SVR algorithm. The following conclusions are obtained:

The decision tree model can identify six VIV modes for the investigated bridge with an accuracy of approximately 94%. The decision tree inputs include the oncoming wind speed and wind direction at three locations on the bridge deck along the span. The “MinLeaf” constraint condition and optimization of the cross-validation method can increase the generalization of the trained model.

The SVR model is proposed to model VIV response. The oncoming wind speed and wind direction at three locations on the bridge deck

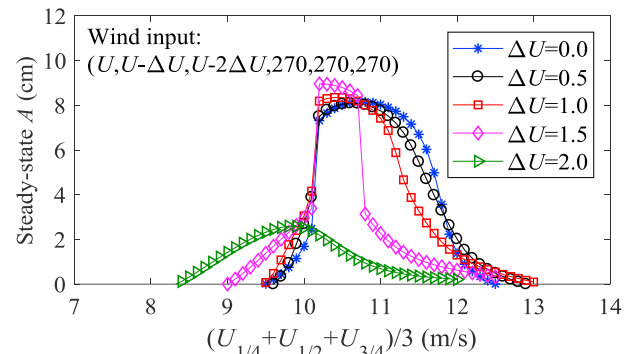


Fig. 18. Influence of nonuniformity on steady-state response.

along the span at the current step, and the response at the previous step are selected as inputs to predict the response at current step. The test results indicate that the SVR model can predict the entire VIV process, including the developing, steady state, and vanishing stages. The VIV response predicted by the SVR model is in good agreement with the field measurement results.

A parametric analysis of VIV response is conducted, using the trained SVR model to map the relationship between wind characteristics and VIV response. The results indicate that when the wind direction is perpendicular to the bridge span, VIV is excited within a certain wind speed range (“lock-in” wind speed range). The developing duration time (from the same initial state to a steady state) is shortest when the wind speed is at the center of the “lock-in” wind speed range, implying intensive interaction between vortices and structural vibration. VIV can occur when the wind direction is within a range of approximately 260° – 280° ; outside this range, the VIV suddenly disappears. The nonuniformity of wind speed along the bridge span has a significant influence on VIV behavior (i.e., the “lock-in” range for the whole bridge becomes narrow as the extent of nonuniformity increases). VIV amplitude is independent from nonuniformity when the extent of nonuniformity is small. VIV amplitude decreases significantly and may even disappear as the extent of nonuniformity further increases. These interesting findings indicate that the ML-based data-driven approach is a powerful tool for wind engineering.

Acknowledgments

This study is financially supported by the National Natural Science Foundation of China under grant Nos. 51638007, 51161120359, 91215302, and 5103138, and the China Postdoctoral Science Foundation under grant No. 2015M580266.

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